

Project Report

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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Platform	ServiceNow (Flow Designer, Service Catalog)

1. INTRODUCTION

1.1 Project Overview

This project demonstrates how the standard laptop ordering process can be automated using ServiceNow Flow Designer. An employee submits a laptop request through the Service Catalog. A flow named “Standard Laptop Task” then processes the request: it obtains approval, creates a Catalog Task for the Hardware team once the request is approved, and sends e-mail notifications to the requester.

The flow is attached to the Standard Laptop catalog item through its Process Engine tab, so every new order is handled automatically with no manual ticket creation. This reduces manual effort, minimises errors and improves the efficiency of IT procurement management.

1.2 Purpose

The purpose of the project is to:

- Give employees a seamless way to request a standard laptop and have it configured on time.
- Remove manual hand-offs and the errors that come with them, such as a configuration task being forgotten.
- Use the IT department’s time better by sending tasks straight to the correct group (Hardware).
- Keep requesters informed at each stage and improve overall productivity of IT procurement operations.

2. IDEATION PHASE

2.1 Problem Statement

The current IT procurement process lacks efficiency and automation, which causes delays and manual overhead, particularly for standard laptop orders. A requested laptop must be configured before the user can use it, but this step is prone to oversight or delay. Requesters are also not told what is happening, which frustrates users and wastes IT resources.

I am	An employee who needs a standard laptop for work
I am trying to	Get a configured laptop quickly through a simple request
But	The request depends on manual follow-up and the configuration task is often missed or delayed
Because	There is no automated approval, task creation or notification process

Which makes me feel	Frustrated, uninformed and unable to start work on time
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2.2 Empathy Map Canvas

The empathy map below is for the main user, the employee who requests a standard laptop. It is based on the problem described in Section 2.1.

SAYS • “When will my laptop be ready?” • “Has my request been approved?” • “I cannot start work without a device.”	THINKS • Is anyone working on my request? • Was my request forgotten? • Why is such a routine request so slow?
DOES • Orders the laptop from the Service Catalog • Checks the request status repeatedly • Follows up with IT by e-mail or phone	FEELS • Frustrated by the waiting • Anxious about missing work deadlines • Uninformed about progress
PAINS • Delays in configuration • No notifications or visibility • Dependence on manual follow-up	GAINS • Laptop configured promptly after approval • Clear e-mail updates at each stage • A simple, one-step request

2.3 Brainstorming

The team considered several ways to solve the problem before selecting one:

S.No.	Idea	Assessment	Decision
1	Continue the manual process and send reminder e-mails to IT staff	Easy to start, but still depends on people and does not remove delays	Rejected
2	Use a legacy ServiceNow Workflow attached to the catalog item	Works, but it is older, harder to maintain and not the current recommended tool	Rejected
3	Write custom scripts / business rules to create tasks	Flexible, but needs coding and is harder for others to maintain	Rejected
4	Use Flow Designer with a Service Catalog trigger, approval, Create Catalog Task and e-mail actions	Low-code, easy to read and change, reusable for other catalog items	Selected

The team selected Idea 4 because it meets all the objectives with the least effort and can be extended easily.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	Request	Approval	Task creation	Configuration	Delivery
User action	Orders Standard Laptop from the Service Catalog	Waits for the approver's decision	No action needed	Waits for the laptop to be configured	Receives the laptop and starts work
Touchpoint	Service Catalog (Order Now), Order Status page	Approval record, e-mail notification	Catalog Task assigned to Hardware	Hardware team, Catalog Task	E-mail, laptop handover
Experience	Simple and quick	Informed of the decision	Instant, with no waiting	Confident that work has started	Satisfied
Pain point (before)	Unsure whether the request was registered	No visibility of the approval	Task created manually, often late	Step forgotten or delayed	Long wait, no update
How the project helps	Request creates REQ and RITM automatically	Approval is requested by the flow	Task is created automatically after approval	Task goes straight to the Hardware group	Requester is notified at each stage

3.2 Solution Requirement

Functional requirements

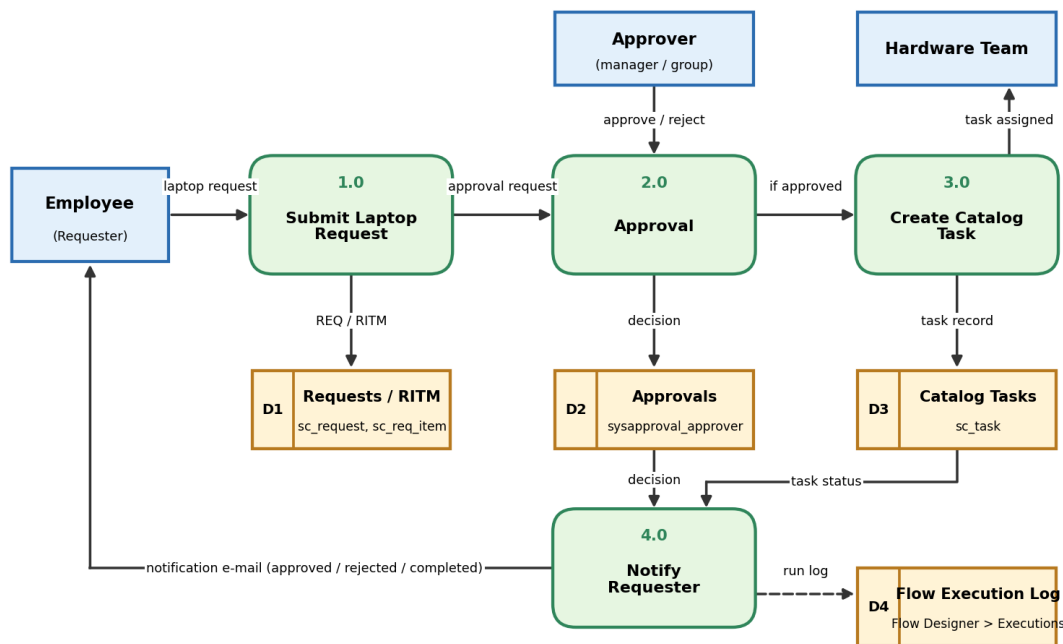
ID	Requirement	Description
FR-1	Catalog ordering	Employees can order the Standard Laptop from Service Catalog > Hardware.
FR-2	Flow trigger	The flow "Standard Laptop Task" starts automatically when the item is ordered (Service Catalog trigger).
FR-3	Approval	The flow asks for approval of the requested item and continues based on the decision.
FR-4	Task creation	If approved, a Catalog Task is created with short description "Laptop needs to be configured" and Assignment group Hardware.
FR-5	Notifications	The requester receives an e-mail when the request is approved or rejected.
FR-6	Flow assignment	The flow is attached to the Standard Laptop item through the Process Engine tab.
FR-7	Traceability	Request (REQ), Requested Item (RITM) and Catalog Task (SCTASK) are linked and every run is logged in Flow Executions.

Non-functional requirements

ID	Requirement	Description
NFR-1	Usability	Ordering must be simple, with no extra steps for the requester.
NFR-2	Reliability	Every approved order must create exactly one task with no manual action.
NFR-3	Security	The flow runs as System User; only users with suitable roles (flow_designer, catalog_admin) can change it.
NFR-4	Maintainability	The flow is low-code and easy to read, so it can be changed without scripting.
NFR-5	Scalability	The same pattern can be reused for other catalog items and groups.
NFR-6	Auditability	Execution history is available in Flow Designer > Executions.

3.3 Data Flow Diagram

Figure 1 shows the Level 1 data flow of the system. The employee submits a request (1.0), the approver decides (2.0), the Catalog Task is created and assigned to the Hardware team (3.0), and the requester is notified (4.0). D1 to D3 are ServiceNow tables, and D4 is the flow execution log.



Level 1 data flow diagram: automated Standard Laptop order

Figure 1: Level 1 data flow diagram of the automated Standard Laptop order

3.4 Technology Stack

Layer	Technology	Purpose
Platform	ServiceNow instance	Hosts the catalog, flow and tables

Layer	Technology	Purpose
Automation	Flow Designer (Process Automation)	Builds the “Standard Laptop Task” flow
Trigger	Service Catalog trigger	Starts the flow when the item is ordered
Actions	Ask for Approval, Create Catalog Task, Send Email	Approval, task creation, notifications
Flow logic	If / Else	Continues only when the request is approved
Catalog setup	Maintain Items > Process Engine tab	Attaches the flow to the Standard Laptop item
Data tables	sc_request, sc_req_item, sc_task, sysapproval_approver	Store request, item, task and approval records
Access	Web browser, roles flow_designer and catalog_admin	Development, ordering and testing

4. PROJECT DESIGN

4.1 Problem Solution Fit

Customer segment	Employees who request standard laptops, the IT procurement team and the Hardware (configuration) team
Problem	Manual follow-up causes missed or delayed configuration, no visibility for the requester and wasted IT effort
Existing alternatives	Manual tickets, e-mail reminders, legacy workflows or custom scripts
Our solution	A Flow Designer flow that approves the request, creates and assigns the Hardware task automatically and notifies the requester
Why it fits	It removes the manual steps that cause the delay, needs no coding and uses the platform the organisation already has
Expected outcome	Faster configuration, fewer errors, better use of IT staff and happier users

4.2 Proposed Solution

Parameter	Description
Problem Statement	Manual handling of standard laptop orders causes delays, missed configuration tasks and poor visibility for the user.
Idea / Solution description	A Flow Designer flow triggered from the Service Catalog requests approval, creates a Catalog Task assigned to the Hardware group once approved, and sends e-mail notifications to the requester.

Parameter	Description
Novelty / Uniqueness	Low-code flow instead of manual tickets or scripts; task creation gated by approval; reusable pattern; full traceability from REQ to RITM to SCTASK.
Social Impact / Customer Satisfaction	Employees get laptops sooner and stay informed; IT staff spend less time on repetitive follow-ups; a consistent, fair process.
Business Model (Revenue Model)	Internal cost-saving solution on the existing ServiceNow subscription; can also be packaged as a reusable template by an implementation partner.
Scalability of the Solution	More catalog items, price-based approvals, location-based groups, technician auto-assignment, asset creation and vendor integration can be added.

4.3 Solution Architecture

Figure 2 shows the architecture and data flow. The requester orders from the Service Catalog, which creates the REQ and RITM records. The Service Catalog trigger starts the flow, which asks the approver for a decision. If the request is approved, the flow creates a Catalog Task assigned to the Hardware team and sends an e-mail to the requester. If it is rejected, only the rejection e-mail is sent. All runs are recorded in the Flow Execution log.

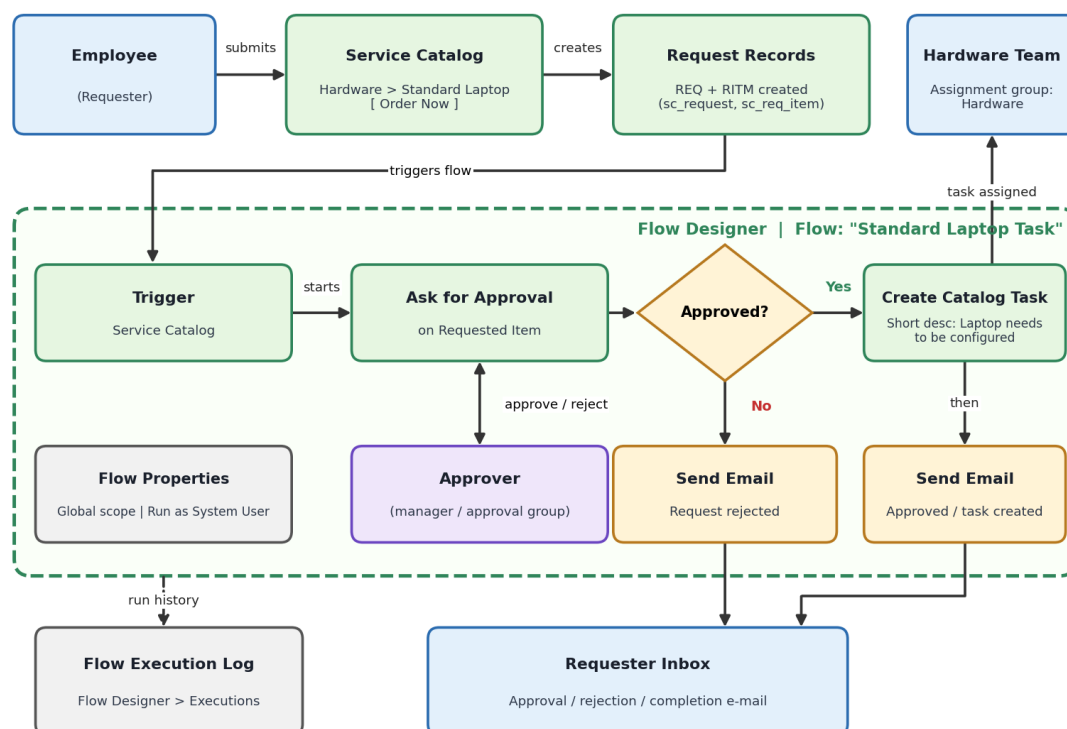


Figure 2: Architecture and data flow of the automated Standard Laptop ordering process in ServiceNow

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned in three milestones that follow the build order.

Milestone	Activities	Deliverable	Status
Milestone 1: Flow	Create the flow "Standard Laptop Task" (Global, run as System User) Add the Service Catalog trigger Add Ask for Approval, If condition, Create Catalog Task and Send Email actions Save and activate the flow	Active flow in Flow Designer	Completed
Milestone 2: Flow assignment	Open Maintain Items > Standard Laptop On the Process Engine tab, clear other automation and select the flow Update the record	Flow attached to the catalog item	Completed
Milestone 3: Service Catalog and testing	Order the Standard Laptop from Service Catalog > Hardware Approve the request from the Approvers tab Verify the Catalog Task, assignment group and notifications Check the flow execution log	Verified end-to-end results	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The solution was tested from both the requester's and the approver's side. The table lists the test cases and what to check in each.

ID	Test case	Expected result	Status
TC-1	Order the Standard Laptop and approve the request	A Catalog Task is created with short description "Laptop needs to be configured" and Assignment group Hardware	Passed (see Section 7.1)
TC-2	Order and reject the approval	No Catalog Task is created and the requester receives a rejection e-mail	To be recorded
TC-3	Check Flow Designer > Executions	Each run shows as completed with every step listed	To be recorded
TC-4	Check the Process Engine tab of the catalog item	The Flow field shows "Standard Laptop Task" and no other automation is selected	Passed (see Section 7.1)

Performance was assessed with the following measures:

Measure	How it is checked	Observation
Task creation	Open the RITM and check the Catalog Tasks tab	The task was created automatically, with no manual step
Assignment accuracy	Check the Assignment group on the task	Assigned to Hardware
Flow run time	Executions log in Flow Designer	To be recorded from the execution log

7. RESULTS

7.1 Output Screenshots

The screenshots below show the main stages of the implementation and the results.

Figure 3: Flow properties (name, application and Run As)

Figure 4: Service Catalog trigger added to the flow

Figure 5: Create Catalog Task action configured with short description and Assignment group Hardware

Figure 6: Flow attached to the Standard Laptop catalog item (Process Engine tab)

Figure 7: Standard Laptop item in the Service Catalog with Order Now

Figure 8: Order status page after the request is submitted

Figure 9: Request record with the Approvers tab

Figure 10: Requested Item showing the automatically created Catalog Task assigned to Hardware

Figure 11: Catalog Task record with short description, description and Approval

8. ADVANTAGES & DISADVANTAGES

Advantages

- Removes manual task creation, so configuration is not forgotten or delayed.
- Tasks reach the Hardware group straight away, which improves use of IT resources.
- Approval comes before work starts, so effort is not spent on rejected requests.
- Requesters are informed, which improves satisfaction.
- Low-code and easy to maintain, and the pattern can be reused for other catalog items.
- Every run can be traced from REQ to RITM to SCTASK and in the execution log.

Disadvantages

- Needs a ServiceNow instance and users with the right roles.
- Covers only standard laptops; purchasing and vendor orders are not yet automated.
- The task is assigned to the Hardware group, not to a specific technician.
- A mistake in the flow or approver setup can stop requests, so changes must be tested.
- The flow depends on the approver responding; no reminder or escalation is included yet.

9. CONCLUSION

Automating standard laptop orders with ServiceNow Flow Designer removes manual hand-offs from the process. Requests are approved, a configuration task is created and assigned to the Hardware team automatically, and users are kept informed. This reduces delays and errors, makes better use of IT staff and gives employees a smoother experience, and it gives the organisation a reusable pattern for other catalog items.

10. FUTURE SCOPE

- Auto-assign the task to a specific technician based on workload or location.
- Create an asset record when the laptop is received.
- Integrate vendor purchase orders and delivery tracking to cover the full procurement cycle.
- Pass the optional software choices (Adobe Acrobat, Adobe Photoshop) and additional requirements into the task description.
- Add reminders and escalation for approvals that stay pending.
- Apply the pattern to other hardware items such as monitors and phones.

11. APPENDIX

Source Code (if any)

No source code was written. The solution is configured in ServiceNow Flow Designer. The main flow settings are:

Setting	Value
Flow name	Standard Laptop Task
Application / Run As	Global / System User
Trigger	Service Catalog
Actions	Ask for Approval; If approved: Create Catalog Task; Send Email
Catalog Task: Short description	Laptop needs to be configured
Catalog Task: Description	Laptop needs to be configured
Catalog Task: Assignment group	Hardware
Attached to	Maintain Items > Standard Laptop > Process Engine > Flow

Dataset Link

Not applicable. The project uses ServiceNow records created during testing and no external dataset.

GitHub & Project Demo Link

GitHub: [add link here]

Project demo video: [add link here]